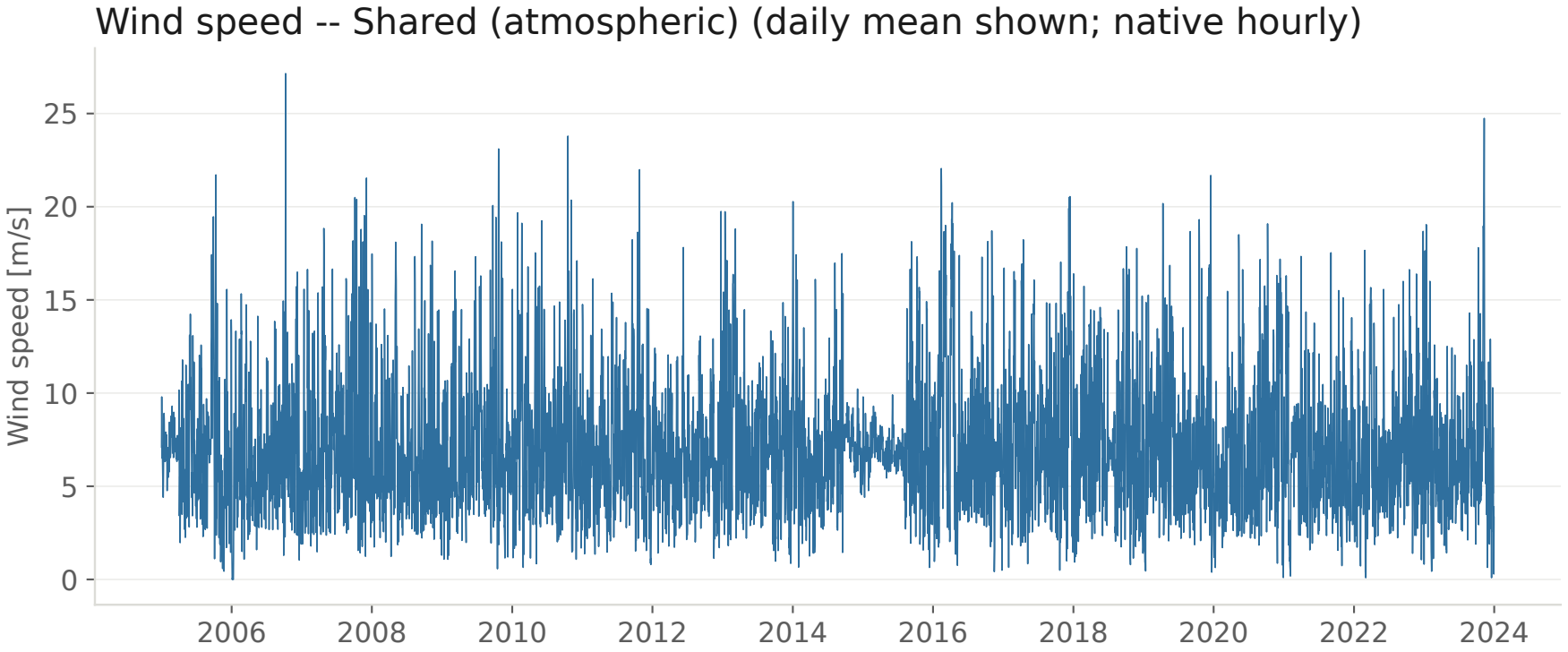
NS-RAD Forcing Report

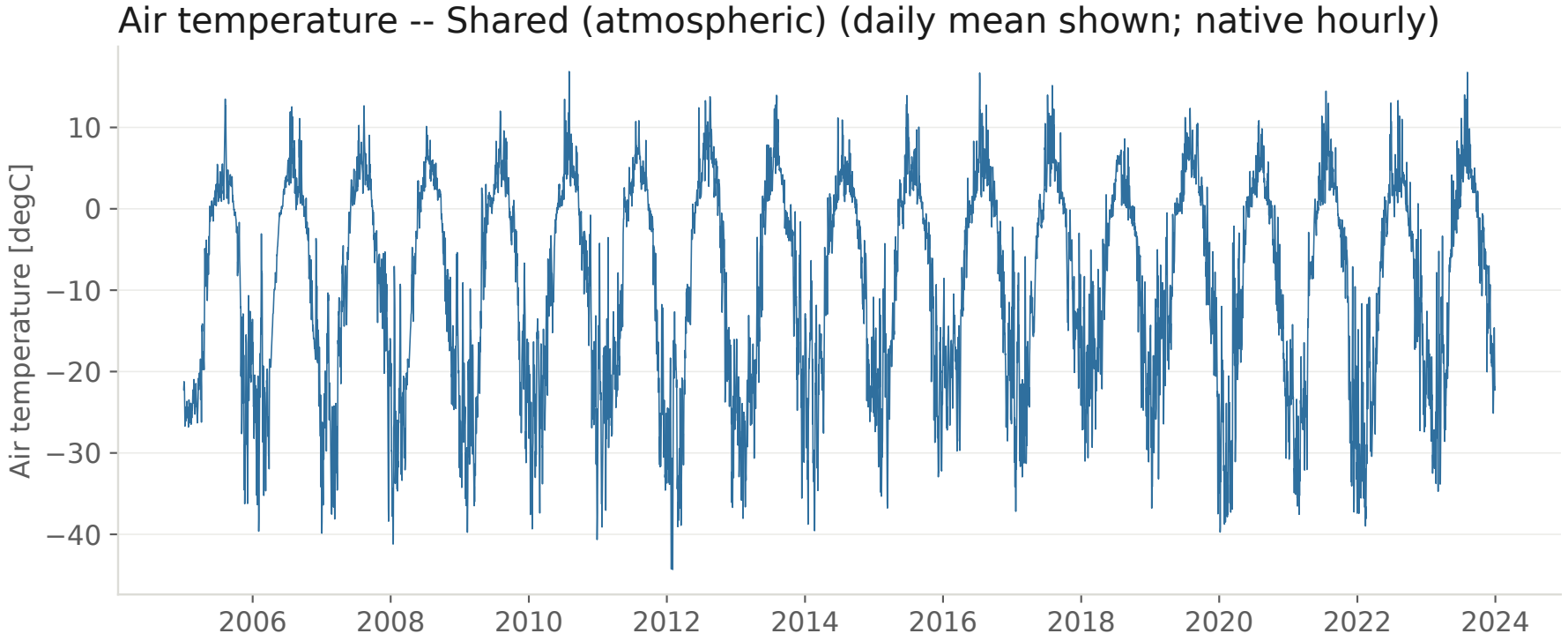
Upstream (riverine) and downstream (marine) boundary forcings,
full-forcing interannual variant, 2005-2023

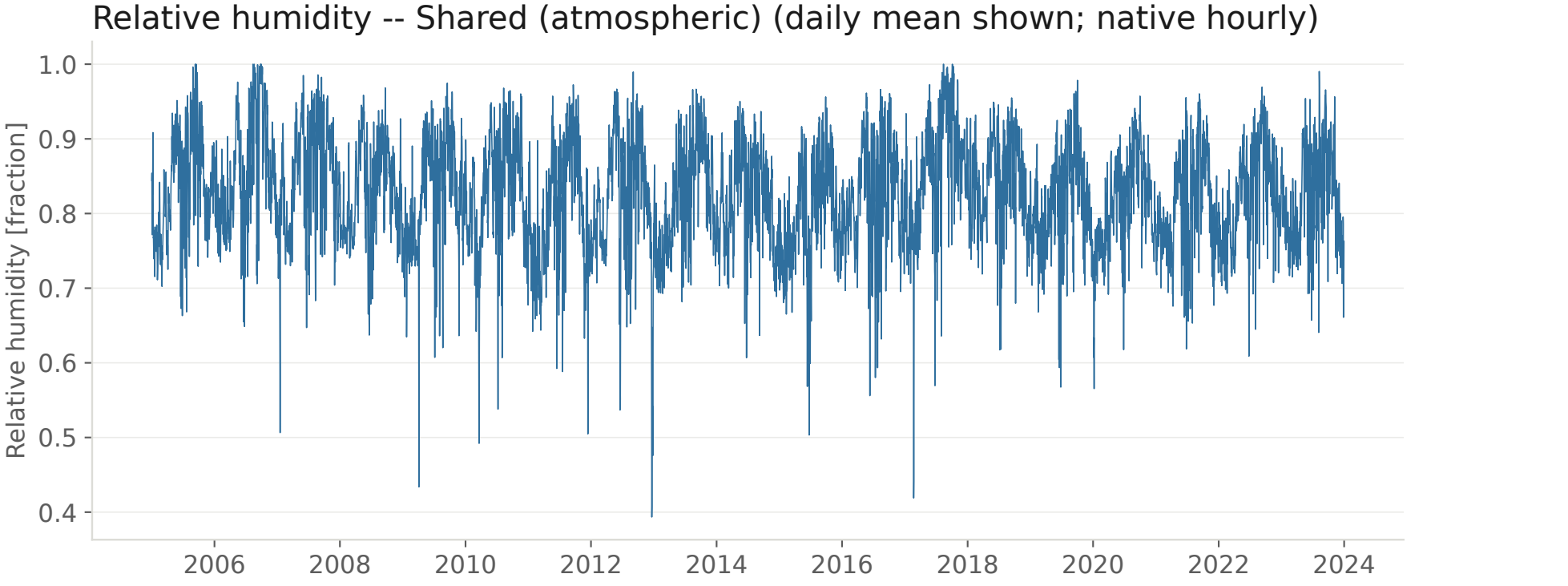
Forcings only -- no model output/state variables are shown here.
See docs/ns_rad_interannual.pdf and docs/publication_figures/ for model output built from these inputs.

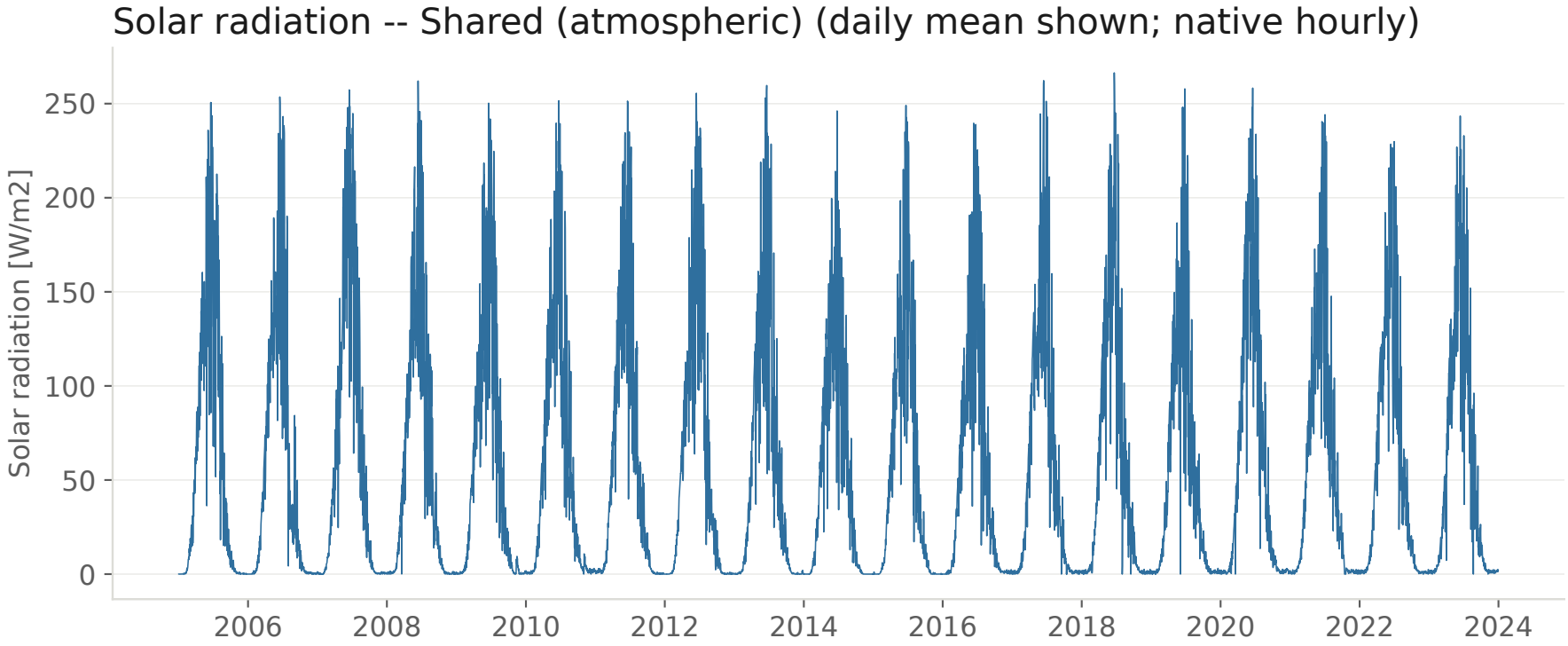
Forcing summary: source and real-data coverage (2005-2023 window)

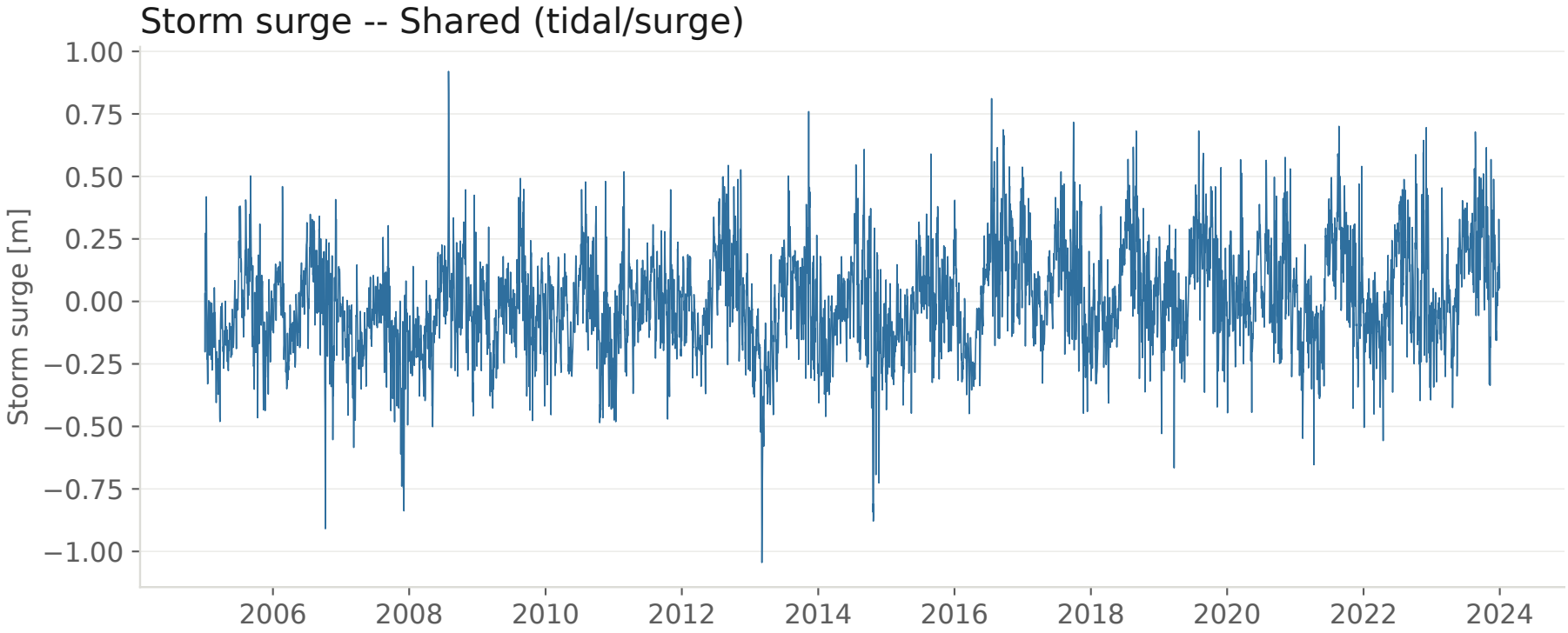
Variable	Category	Source	Native frequency	Period coverage	% real	Units
Wind speed	Shared (atmospheric)	NDBC PRDA2, Prudhoe Bay	Hourly (native)	2005-2023 (station start)	89%	m/s
Air temperature	Shared (atmospheric)	NDBC PRDA2, Prudhoe Bay	Hourly (native)	2005-2023 (station start)	93%	degC
Relative humidity	Shared (atmospheric)	Deadhorse Airport (NOAA ISD)	Hourly (native)	2005-2023 (real from 1980)	95%	fraction
Solar radiation	Shared (atmospheric)	NOAA GML Barrow (Utqiagvik) BSRN	1-minute (native) -> hourly mean	2005-2023 (real since 1998)	98%	W/m2
Storm surge	Shared (tidal/surge)	NOAA CO-OPS 9497645, Prudhoe Bay	Hourly -> daily mean (tide must cancel out)	2005-2023 (real since 1995)	99%	m
Riverine temperature	Upstream (riverine)	Derived (air->water formula)	Derived at daily resolution (from daily air temp)	2005-2023	n/a (derived)	degC
River discharge	Upstream (riverine)	PWBM	Daily (native -- PWBM's own daily output)	1980-2023 (sliced)	100%	m3/s
Upstream DOC (TOC, cub)	Upstream (riverine)	PWBM	Daily (native -- PWBM's own daily output)	1980-2023 (sliced)	100%	mmol C/m3
Salinity	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	PSU
Sea temperature	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (was PRDA2)	100%	degC
DIC	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	mmol C/m3
Alkalinity	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	mmol/m3
Nitrate (NO3)	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	mmol N/m3
Ammonium (NH4)	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	mmol N/m3
Phosphate (PO4)	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	mmol P/m3
Dissolved silica (dSi)	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	mmol Si/m3
Dissolved oxygen	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	mmol O2/m3
Marine DOC (TOC, clb)	Downstream (marine)	ECCO-Darwin v5 (extended)	Monthly (native) -> interpolated to daily	1995-2023 (sliced)	100%	mmol C/m3
Tides (harmonic)	Downstream (marine)	NOAA CO-OPS (per-river station)	Continuous (analytic formula)	unlimited (analytic)	100%	m

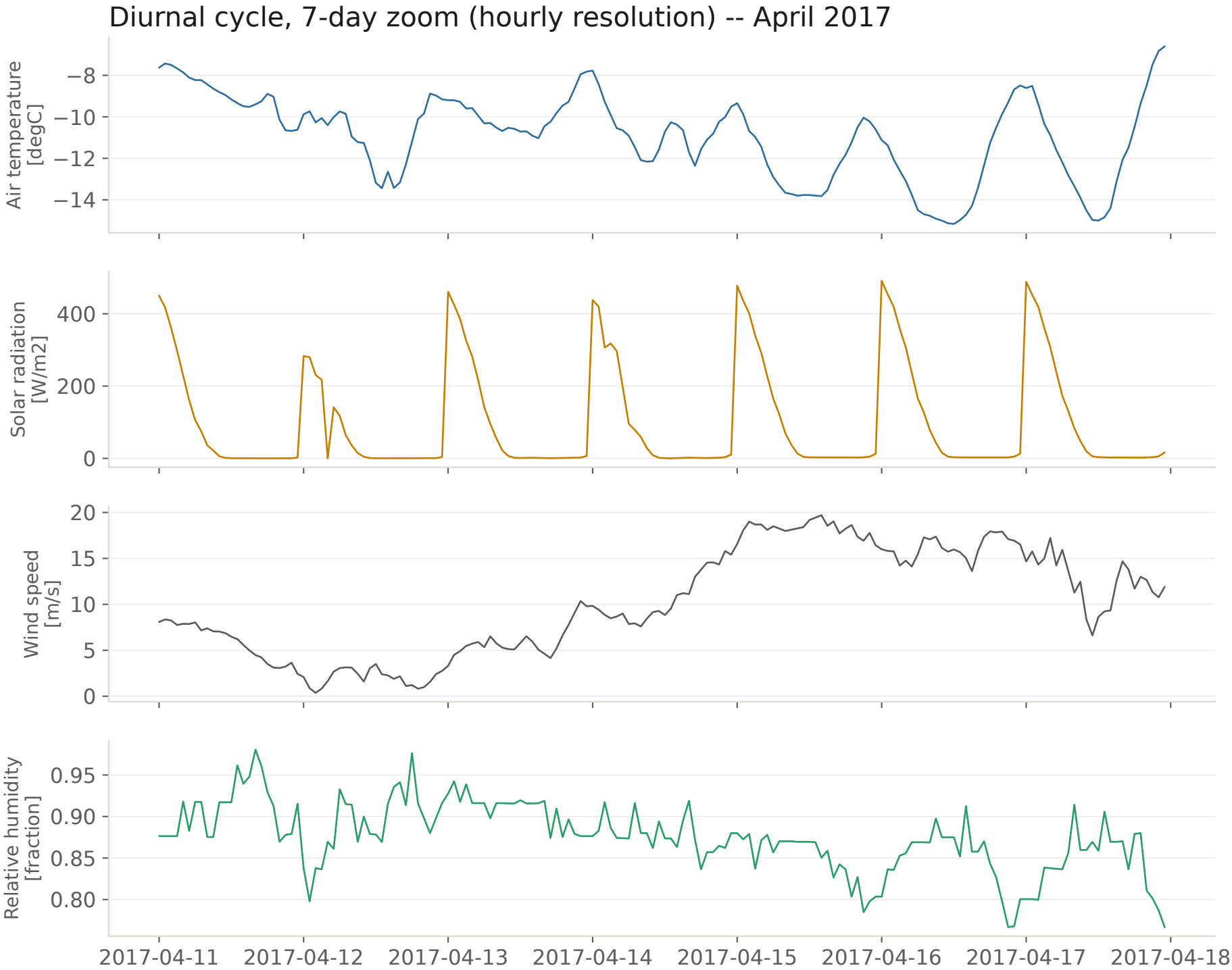

Wind speed -- Shared (atmospheric) (daily mean shown; native hourly)
Wind speed [m/s]
0
5
10
15
20
25
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · NDBC PRDA2, Prudhoe Bay · native frequency: Hourly (native)

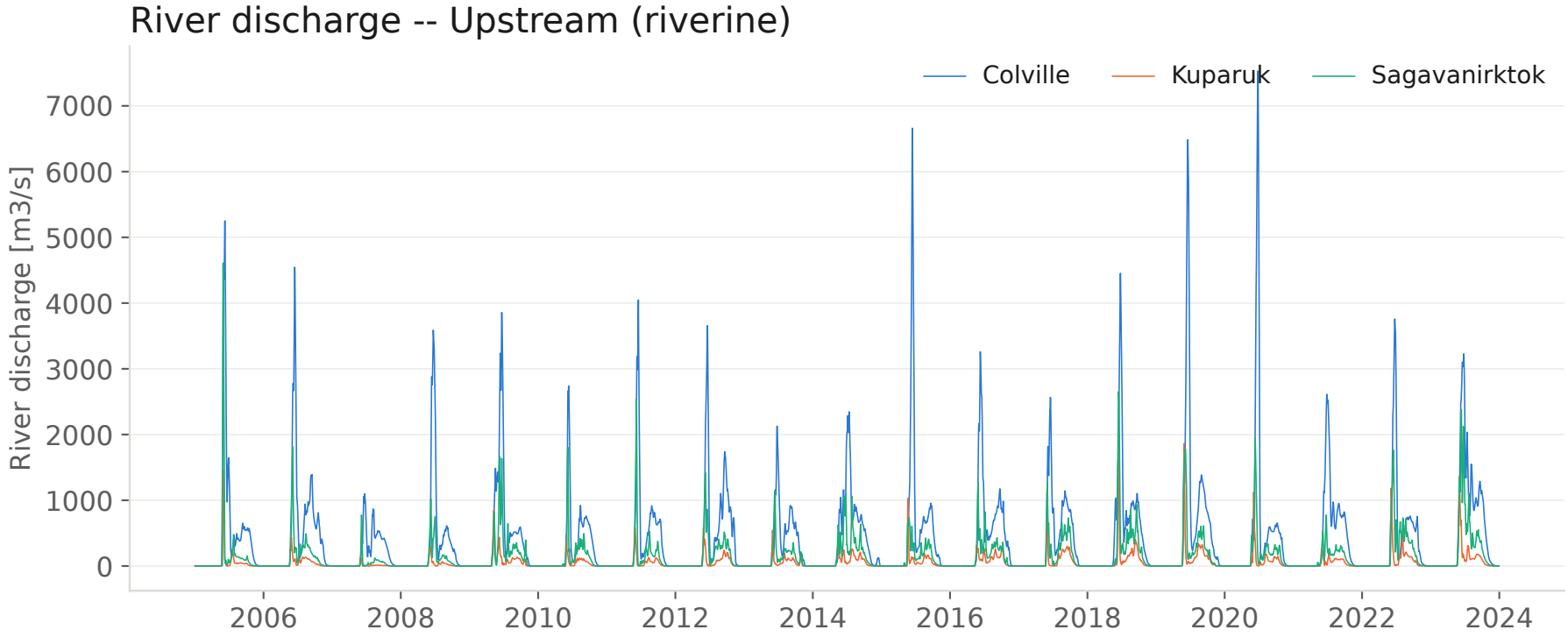

Air temperature -- Shared (atmospheric) (daily mean shown; native hourly)
Air temperature [degC]
10
0
−10
−20
−30
−40
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · NDBC PRDA2, Prudhoe Bay · native frequency: Hourly (native)


Relative humidity -- Shared (atmospheric) (daily mean shown; native hourly)
Relative humidity [fraction]
1.0
0.9
0.8
0.7
0.6
0.5
0.4
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · Deadhorse Airport (NOAA ISD) · native frequency: Hourly (native)

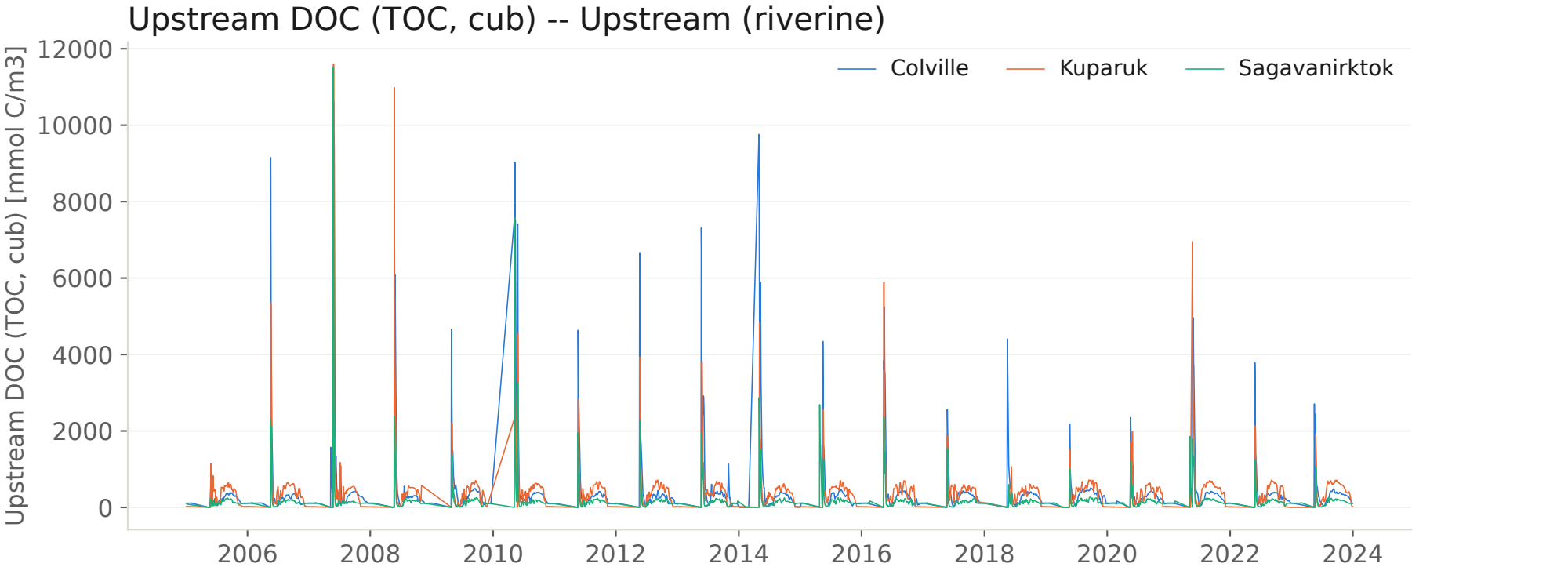

Solar radiation -- Shared (atmospheric) (daily mean shown; native hourly)
Solar radiation [W/m2]
250
200
150
100
50
0
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · NOAA GML Barrow (Utqiagvik) BSRN · native frequency: 1-minute (native) -> hourly mean

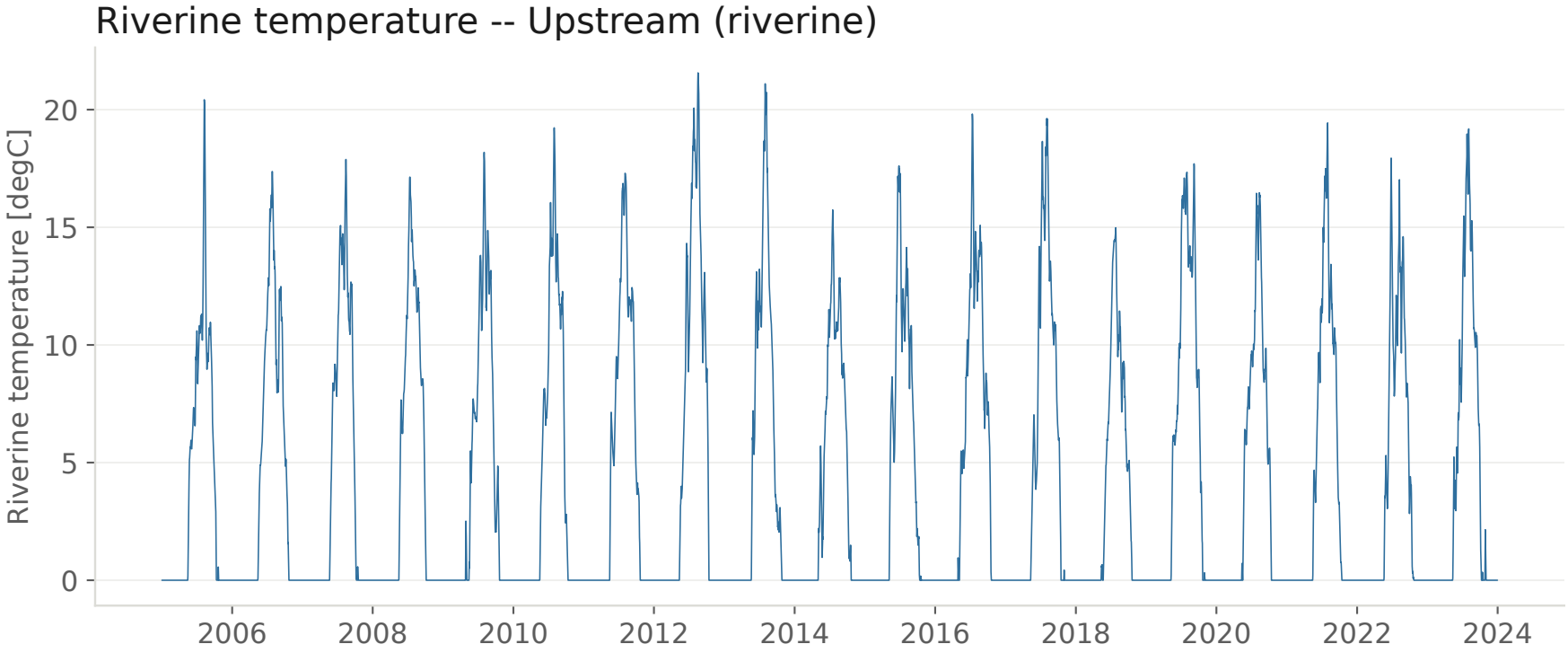

Storm surge -- Shared (tidal/surge)
Storm surge [m]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · NOAA CO-OPS 9497645, Prudhoe Bay · native frequency: Hourly -> daily mean (tide must cancel out)

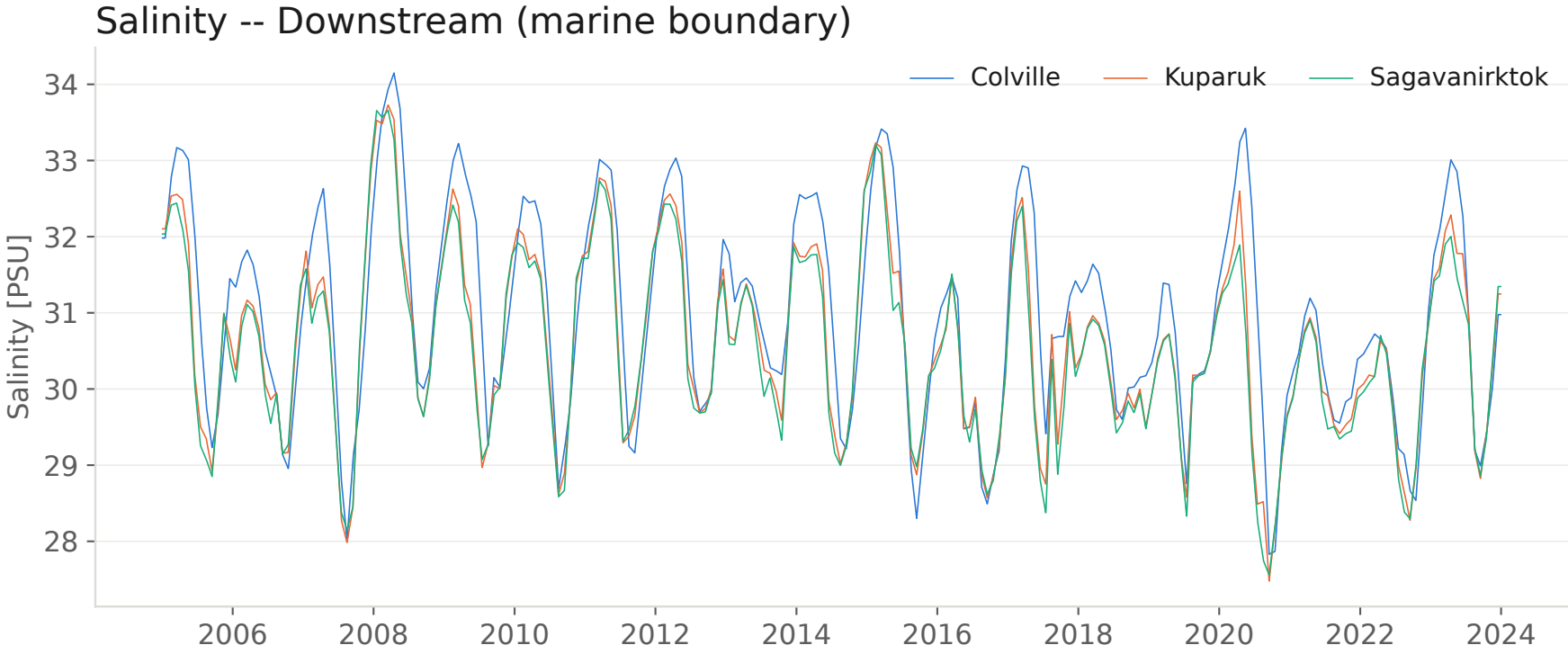

Diurnal cycle, 7-day zoom (hourly resolution) -- April 2017
Air temperature [degC]
−8
−10
−12
−14
Solar radiation [W/m2]
400
200
0
Wind speed [m/s]
20
15
10
5
0
Relative humidity [fraction]
0.95
0.90
0.85
0.80
2017-04-11
2017-04-12
2017-04-13
2017-04-14
2017-04-15
2017-04-16
2017-04-17
2017-04-18

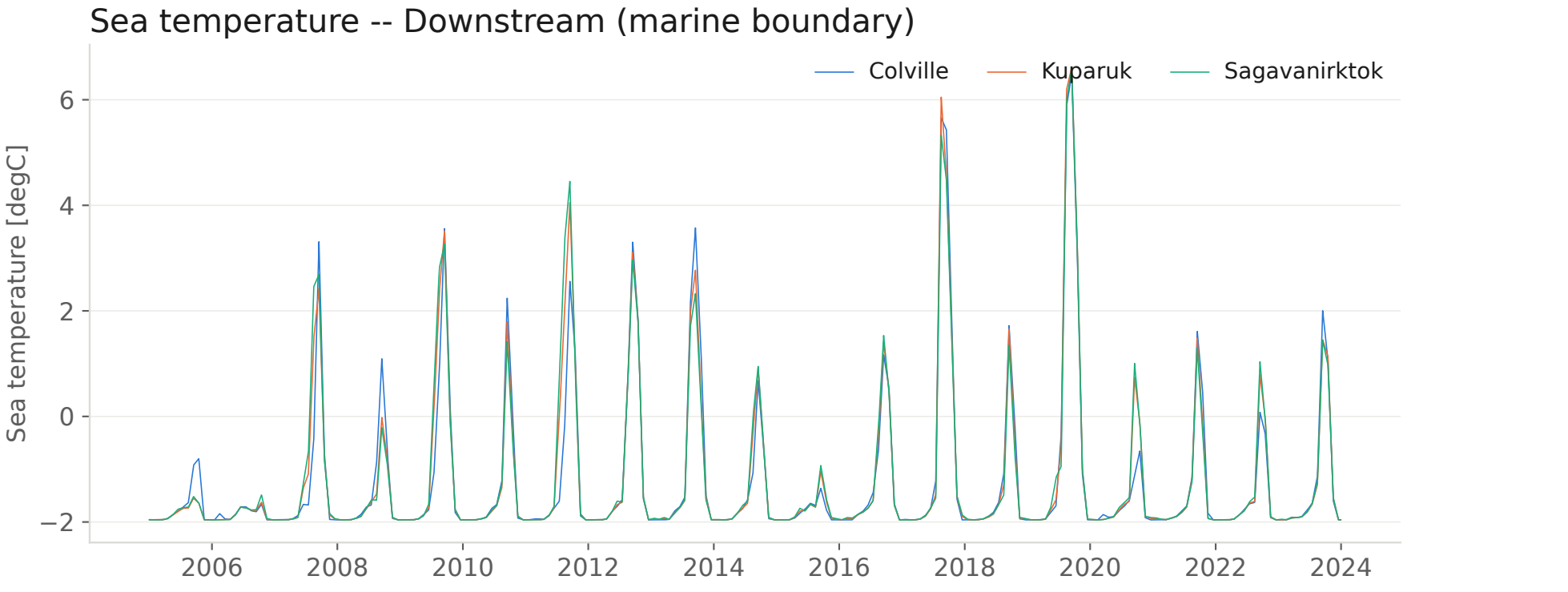


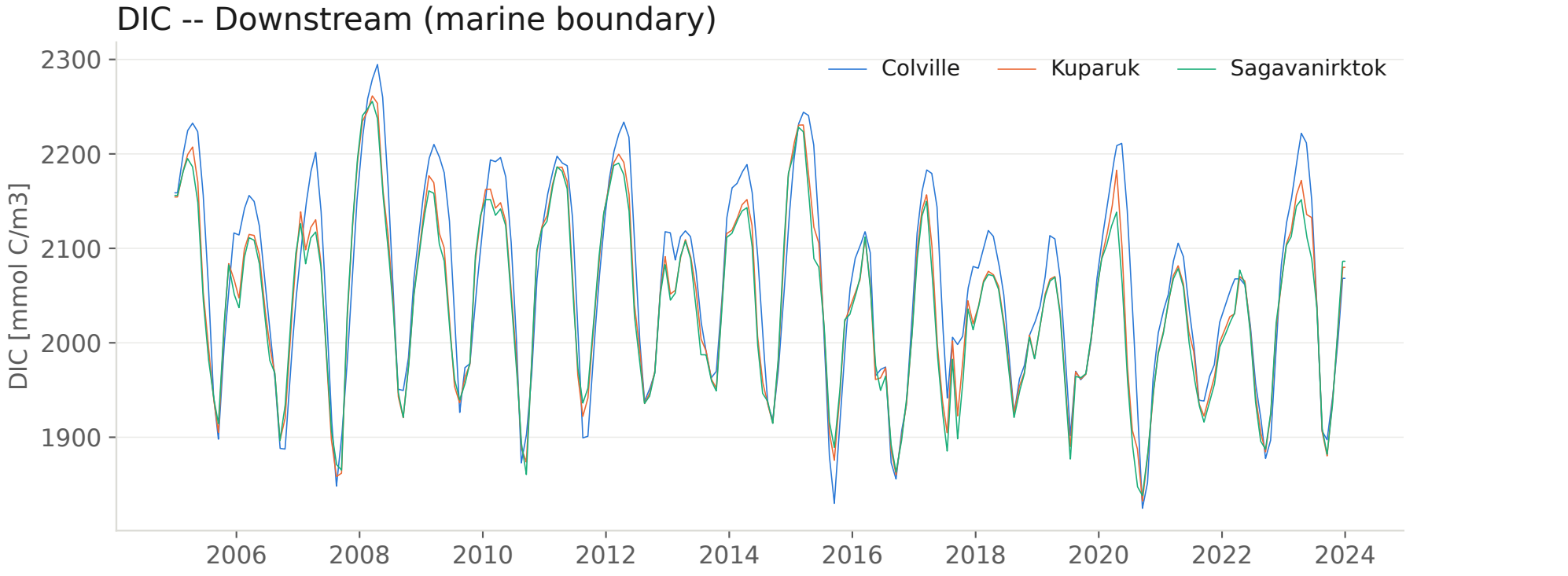
NS-RAD — North Slope River-Aquatic-Delta Model · PWBM · native frequency: Daily (native -- PWBM's own daily output)

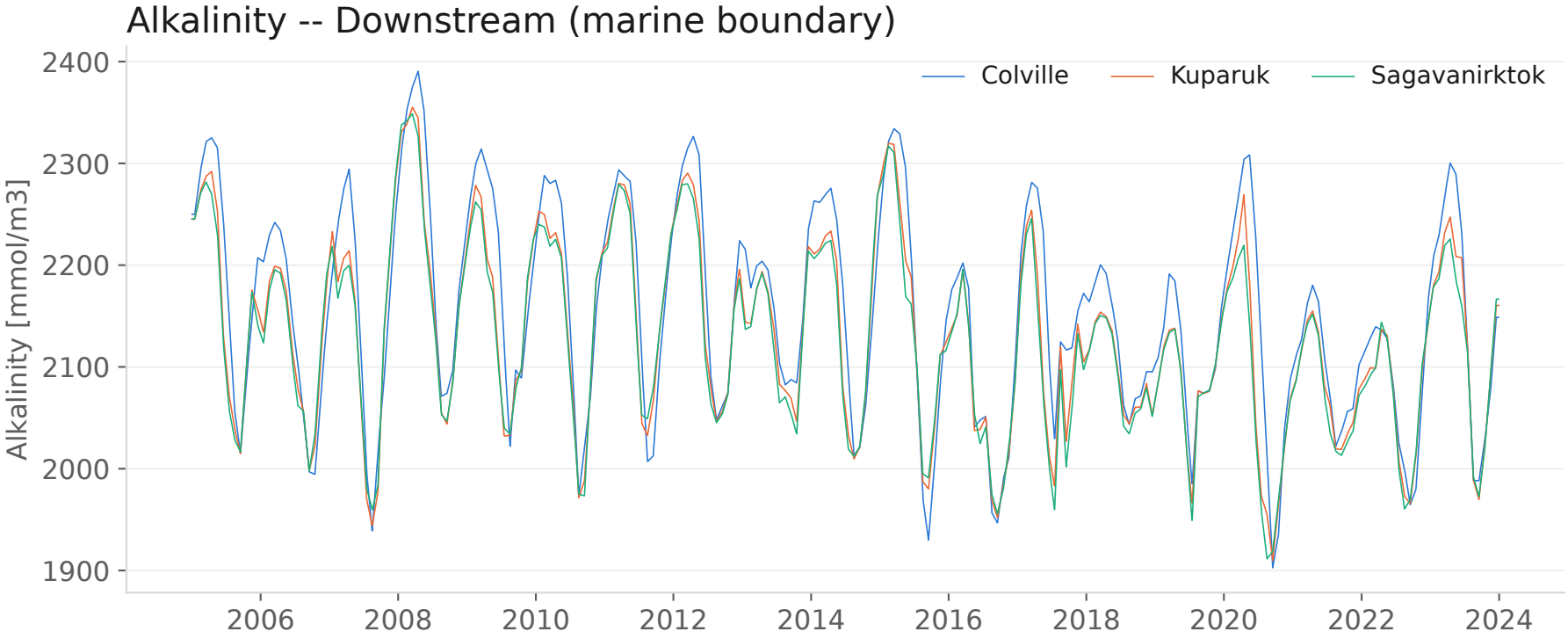

Upstream DOC (TOC, cub) -- Upstream (riverine)
Upstream DOC (TOC, cub) [mmol C/m3]
12000
10000
8000
6000
4000
2000
0
Colville
Kuparuk
Sagavanirktok
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · PWBM · native frequency: Daily (native -- PWBM's own daily output)

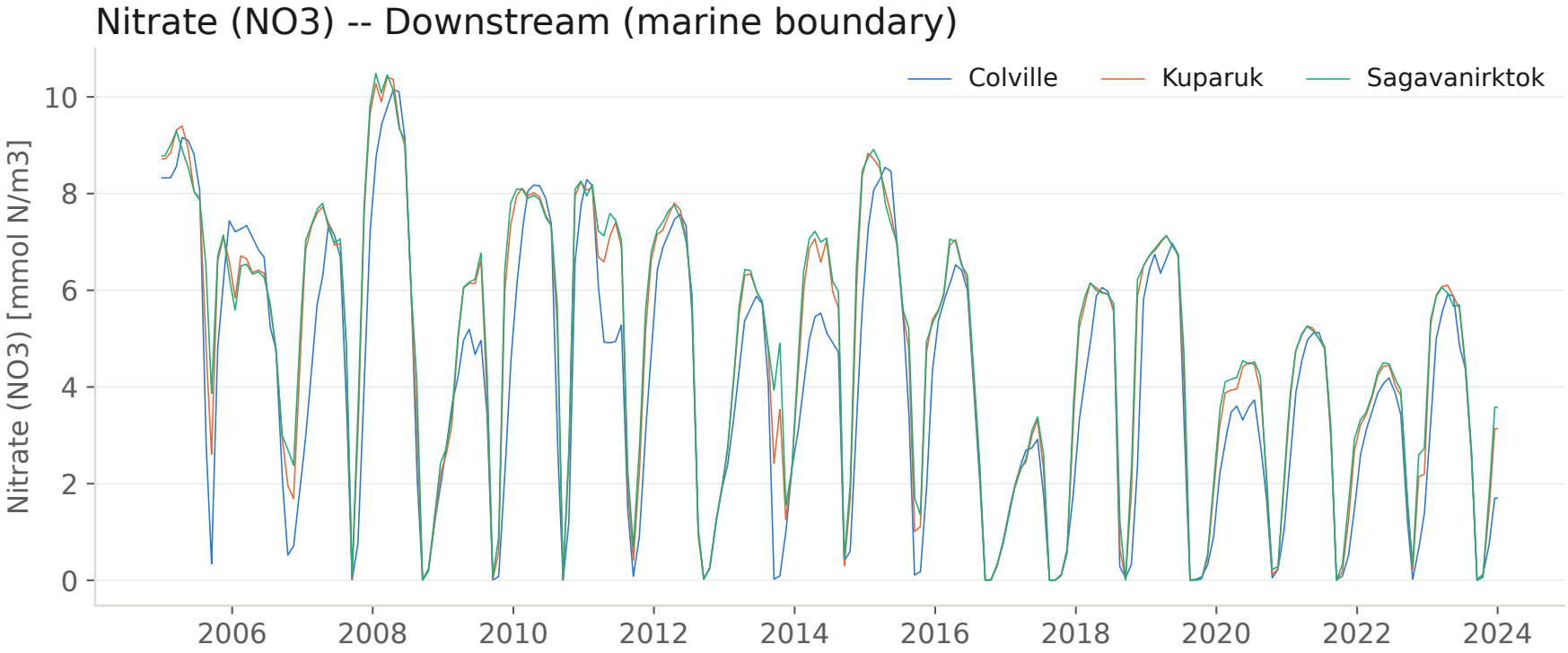

Riverine temperature -- Upstream (riverine)
Riverine temperature [degC]
0
5
10
15
20
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · Derived (air->water formula) · native frequency: Derived at daily resolution (from daily air temp)

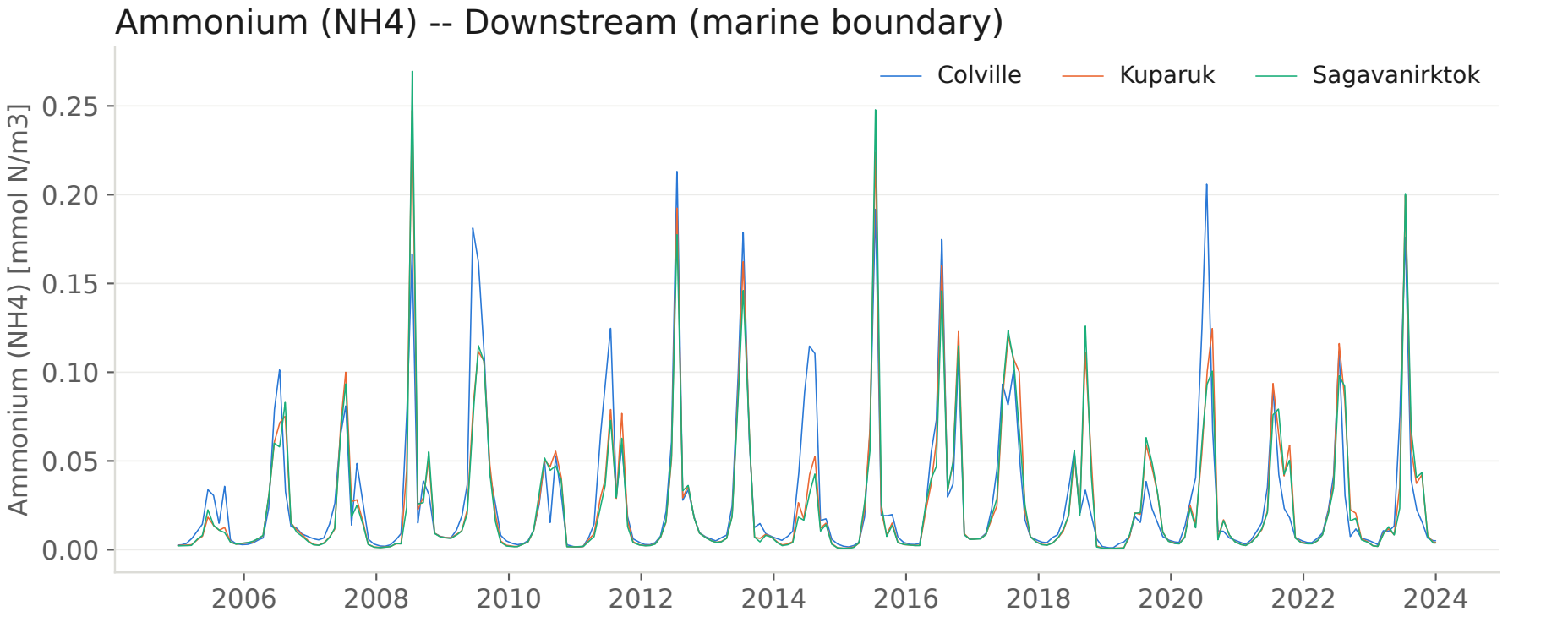

Salinity -- Downstream (marine boundary)
Colville
Kuparuk
Sagavanirktok
Salinity [PSU]
34
33
32
31
30
29
28
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily

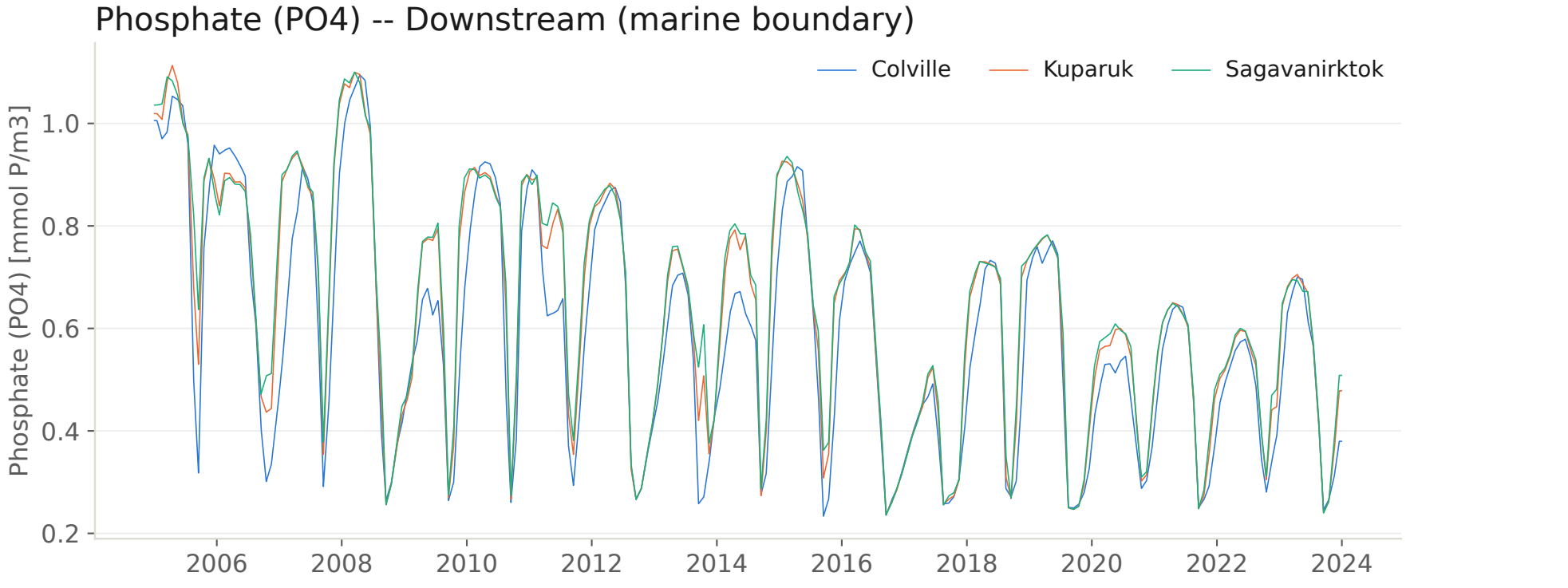

Sea temperature -- Downstream (marine boundary)
Colville
Kuparuk
Sagavanirktok
Sea temperature [degC]
6
4
2
0
−2
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily

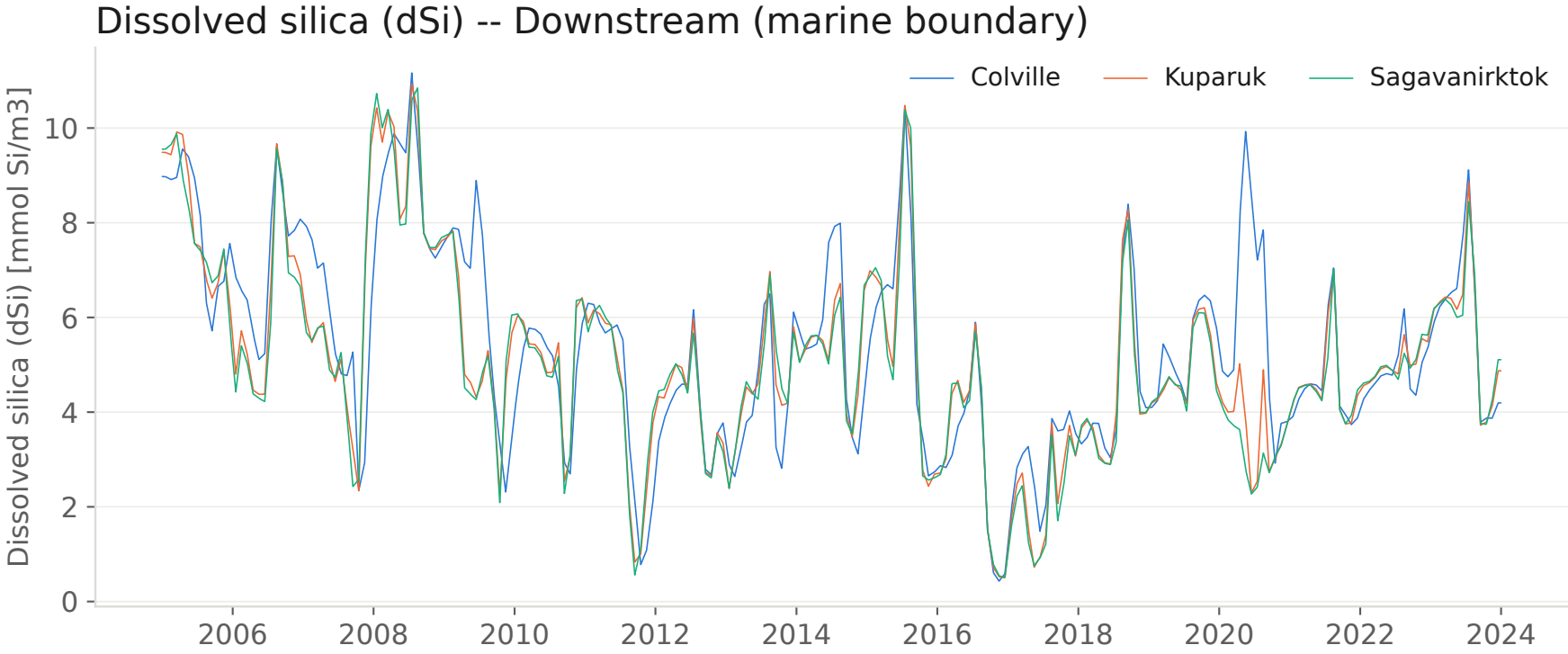

DIC -- Downstream (marine boundary)
DIC [mmol C/m3]
2300
2200
2100
2000
1900
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Colville
Kuparuk
Sagavanirktok
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily

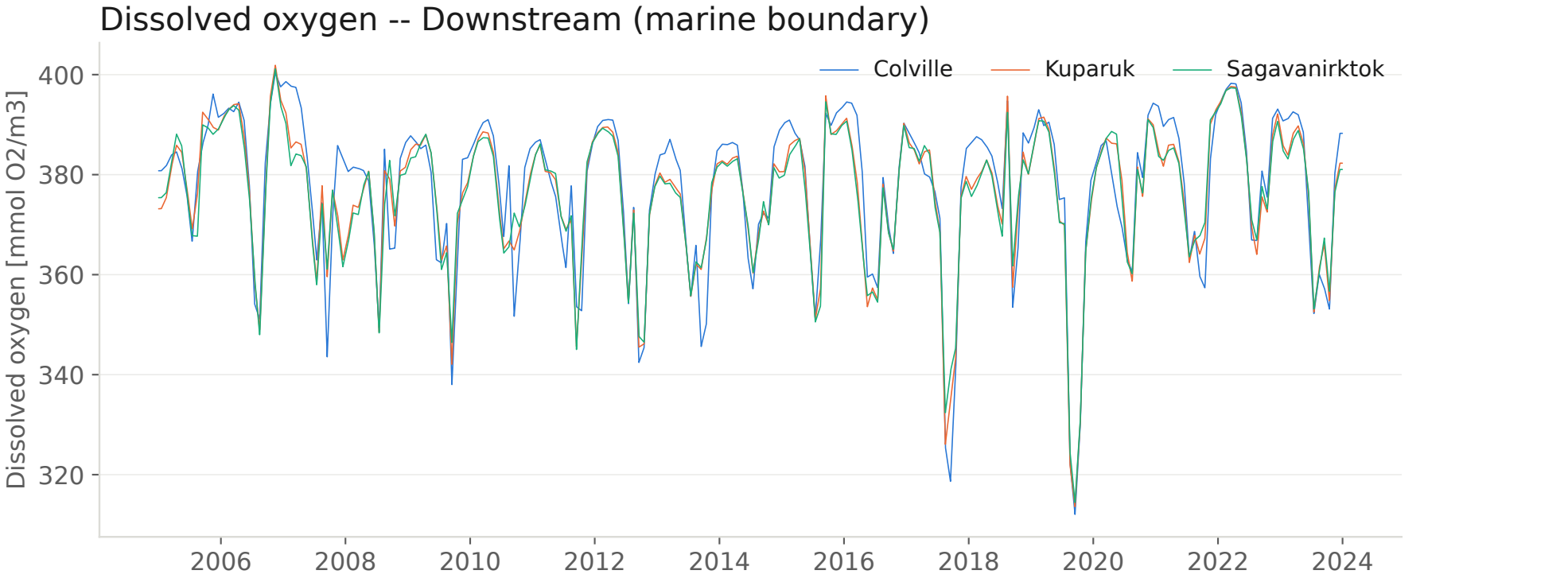

Alkalinity -- Downstream (marine boundary)
Alkalinity [mmol/m3]
2400
2300
2200
2100
2000
1900
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Colville
Kuparuk
Sagavanirktok
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily

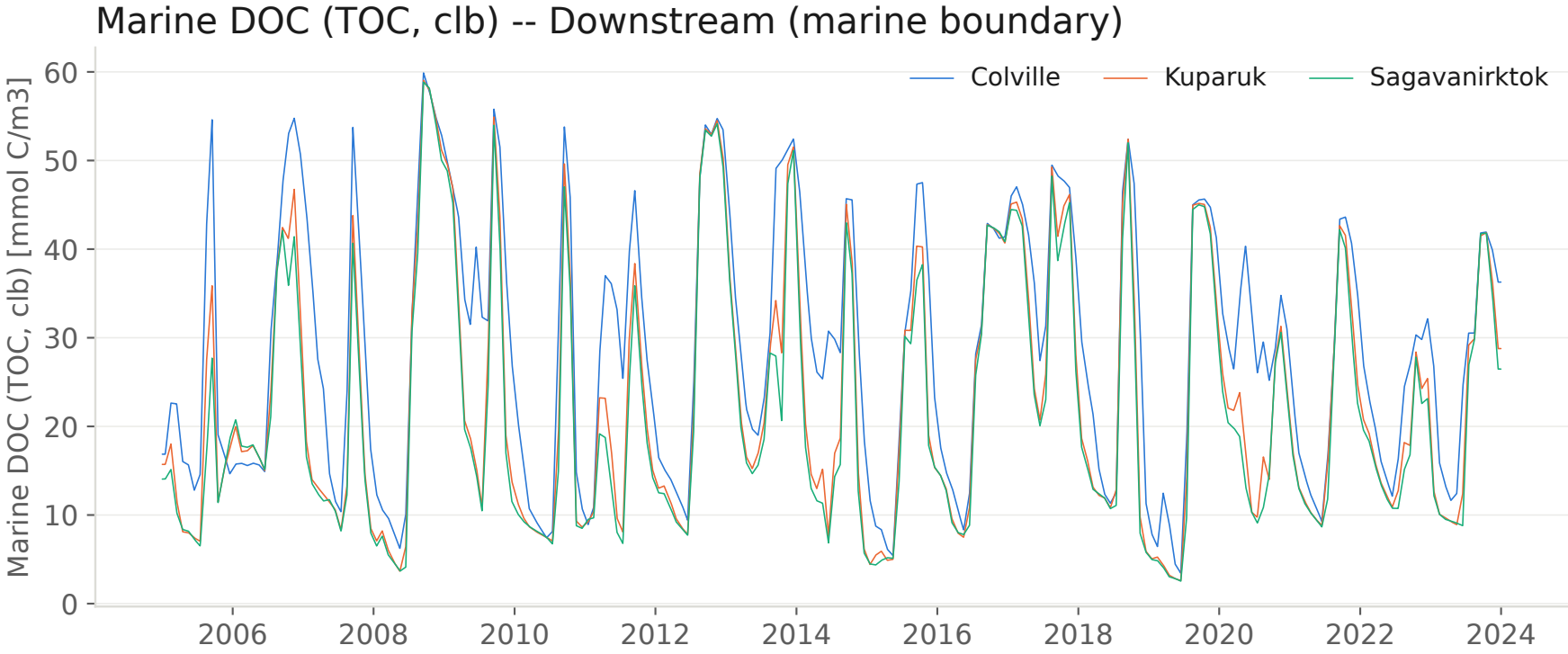

Nitrate (NO3) -- Downstream (marine boundary)
Colville
Kuparuk
Sagavanirktok
Nitrate (NO3) [mmol N/m3]
0
2
4
6
8
10
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily


Ammonium (NH4) -- Downstream (marine boundary)
Colville
Kuparuk
Sagavanirktok
Ammonium (NH4) [mmol N/m3]
0.25
0.20
0.15
0.10
0.05
0.00
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily

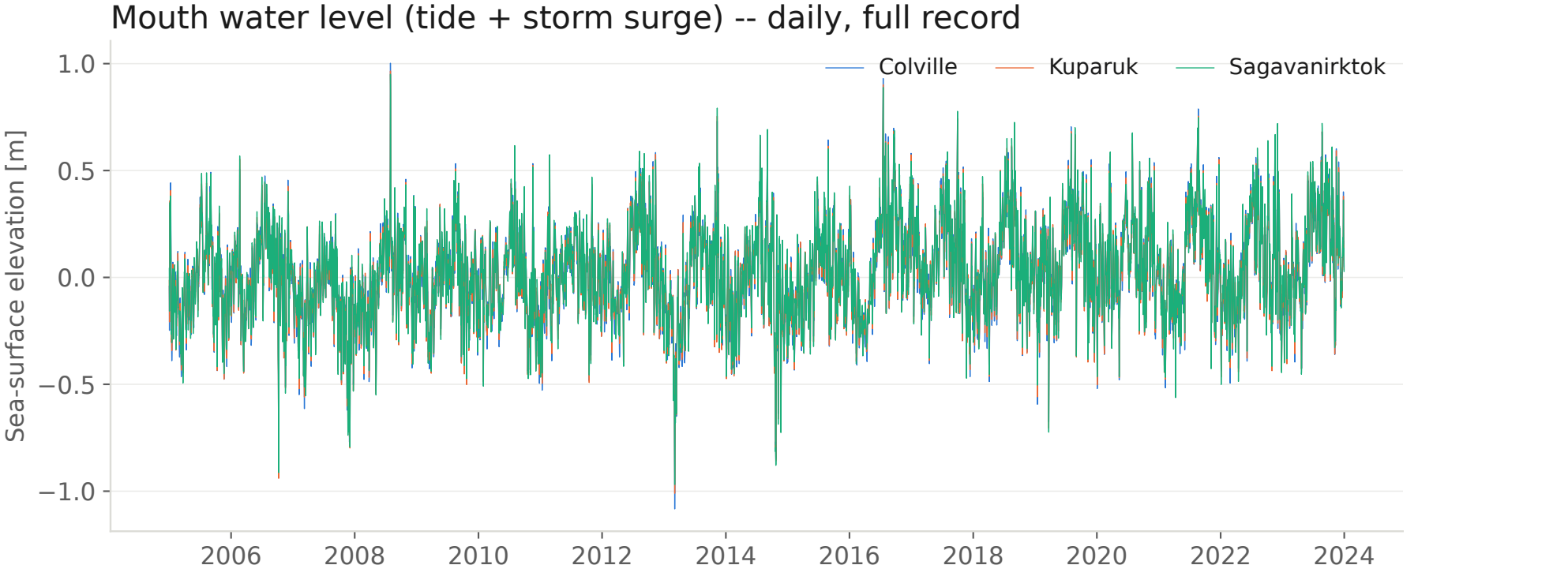

Phosphate (PO4) -- Downstream (marine boundary)
Colville
Kuparuk
Sagavanirktok
Phosphate (PO4) [mmol P/m3]
1.0
0.8
0.6
0.4
0.2
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily

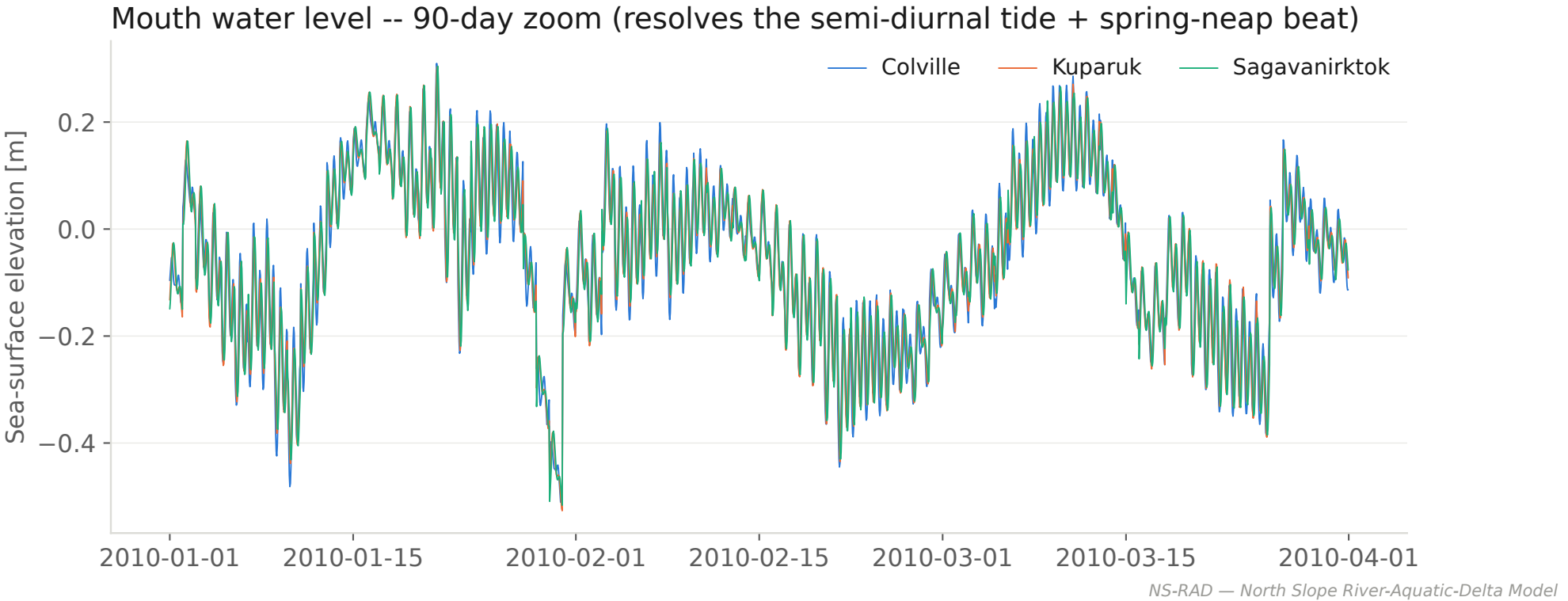

Dissolved silica (dSi) -- Downstream (marine boundary)
Colville
Kuparuk
Sagavanirktok
Dissolved silica (dSi) [mmol Si/m3]
0
2
4
6
8
10
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily


Dissolved oxygen -- Downstream (marine boundary)
Colville
Kuparuk
Sagavanirktok
Dissolved oxygen [mmol O2/m3]
400
380
360
340
320
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily



NS-RAD — North Slope River-Aquatic-Delta Model · ECCO-Darwin v5 (extended) · native frequency: Monthly (native) -> interpolated to daily


Mouth water level (tide + storm surge) -- daily, full record
Colville
Kuparuk
Sagavanirktok
Sea-surface elevation [m]
1.0
0.5
0.0
−0.5
−1.0
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
NS-RAD — North Slope River-Aquatic-Delta Model · Harmonic tide (continuous, analytic) + Prudhoe Bay surge proxy (hourly obs -> daily mean)


Mouth water level -- 90-day zoom (resolves the semi-diurnal tide + spring-neap beat)
Colville
Kuparuk
Sagavanirktok
Sea-surface elevation [m]
0.2
0.0
−0.2
−0.4
2010-01-01
2010-01-15
2010-02-01
2010-02-15
2010-03-01
2010-03-15
2010-04-01
NS-RAD — North Slope River-Aquatic-Delta Model